gpeyt1-01 · SPT (N1)60

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A boring for a highway embankment encounters 8 ft of moist silty sand (γ = 112 pcf) over saturated clean fine sand (γsat = 124 pcf). The water table is at 8 ft. At a depth of 22 ft the field SPT blow count is N = 14 blows/ft. The automatic hammer was calibrated at an energy ratio of 80%. The driller also reports a 2 ft sample recovery, a 140 lb hammer with a 30 in. drop, and AWJ rods (none of which need further correction for this problem).

Most nearly, the corrected blow count (N1)60 at 22 ft is:

A. 20
B. 17
C. 15
D. 11

Given equations / data

> γw = 62.4 pcf; N60 = (Eeff/60)·Nmeas; (N1)60 = CN·N60; CN = 0.77 log10(20/Po), Po = vertical effective stress in tsf

Worked solution

1. Po = [8(112) + 14(124 − 62.4)]/2000 = 0.879 tsf
2. N60 = 14×80/60 = 18.67
3. CN = 0.77 log10(20/0.879) = 1.045
4. (N1)60 = 18.67×1.045 = 19.5 ≈ 20

Final answer

20

Why the other values are wrong

- **17** — used total stress for Po
- **15** — skipped the energy correction
- **11** — inverted the energy ratio (60/80)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2 (Hammer Efficiency, Depth Adjustment), p. 124

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwQ24-zlxyU, https://www.youtube.com/watch?v=DJaHlgyaOAg (no creator values copied).

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gpeyt1-02 · RQD

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A 5.0 ft (60 in.) NQ core run in a jointed limestone recovers the following intact pieces (in., measured along the core axis): 5.5, 2.0, 7.0, 3.5, 4.0, 1.5, 9.0, 3.0, 6.5, 2.5, 4.5. The rest of the run was lost as rubble. The core barrel was double-tube, drilling took 18 minutes, and the drill water return was 80%.

Most nearly, the RQD of this run and its rock-quality description are:

A. 82%, good
B. 54%, fair
C. 74%, fair
D. 61%, fair

Given equations / data

> RQD = Σ(length of intact pieces ≥ 4 in.)/(total core run length) × 100%

Worked solution

1. Pieces ≥ 4 in.: 5.5, 7.0, 4.0, 9.0, 6.5, 4.5 → Σ = 36.5 in.
2. RQD = 36.5/60 × 100 = 60.8%
3. Recovery = 49.0/60 = 82% (not RQD)
4. RQD 61% → fair (handbook RQD description table)

Final answer

61%, fair

Why the other values are wrong

- **82%, good** — reported total core recovery (all pieces) as RQD
- **54%, fair** — excluded pieces exactly 4 in. long
- **74%, fair** — divided by recovered length instead of run length

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.4.1 Rock Quality Designation (RQD), p. 120

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gpeyt1-03 · USCS from gradation

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sieve analysis on a split-spoon sample gives: 96% passing No. 4, 3% passing No. 200, D10 = 0.12 mm, D30 = 0.3 mm, D60 = 0.95 mm. The sample is non-plastic, brown, moist, with a natural water content of 9%. The boring log also notes trace mica.

Most nearly, Cu, Cc and the USCS group symbol are:

A. Cu = 7.9, Cc = 1.27, SW
B. Cu = 7.9, Cc = 0.79, SP-SM
C. Cu = 7.9, Cc = 0.79, SP
D. Cu = 7.9, Cc = 0.79, SW

Given equations / data

> Cu = D60/D10; Cc = D30²/(D10·D60); SW requires Cu ≥ 6 and 1 ≤ Cc ≤ 3; fines < 5%

Worked solution

1. Cu = 0.95/0.12 = 7.92 (≥ 6 OK)
2. Cc = 0.3²/(0.12×0.95) = 0.789 (< 1, fails)
3. 96% passes No. 4 → sand; 3% fines < 5% → clean sand
4. Fails Cc → poorly graded sand, SP

Final answer

Cu = 7.9, Cc = 0.79, SP

Why the other values are wrong

- **Cu = 7.9, Cc = 1.27, SW** — inverted Cc
- **Cu = 7.9, Cc = 0.79, SP-SM** — treated 3% fines as a dual-symbol (5–12%) soil
- **Cu = 7.9, Cc = 0.79, SW** — checked Cu only

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.7.2 USCS, p. 110–117; §3.8.4 Gradation Tests, p. 136

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gpeyt1-04 · Atterberg / LI

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A Shelby-tube sample of gray clay from 30 ft has LL = 58, PL = 24, and natural water content w = 52%. The torvane reading was 0.35 tsf, the pocket penetrometer 0.5 tsf, and 100% passes the No. 200 sieve. The engineer wants the liquidity index and a USCS fine-grained symbol from the plasticity chart.

Most nearly, the LI and USCS symbol are:

A. LI = 0.18, CH
B. LI = 0.82, MH
C. LI = 0.48, CL
D. LI = 0.82, CH

Given equations / data

> PI = LL − PL; LI = (w − PL)/PI; A-line: PI = 0.73(LL − 20)

Worked solution

1. PI = 58 − 24 = 34
2. LI = (52 − 24)/34 = 0.824
3. A-line at LL = 58: 0.73(58−20) = 27.7 < PI = 34 → above A-line
4. LL ≥ 50 and above A-line → CH; LI near 0.9 → soft, near-liquid consistency

Final answer

LI = 0.82, CH

Why the other values are wrong

- **LI = 0.18, CH** — computed consistency index (LL − w)/PI
- **LI = 0.82, MH** — read the soil as below the A-line
- **LI = 0.48, CL** — divided by LL and used LL < 50 boundary

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.2 Atterberg Limits, p. 132; §3.7.2 plasticity chart

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gpeyt1-05 · (N1)60 profile

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A sand deposit has γ = 118 pcf above the water table at 6 ft and γsat = 128 pcf below. Two SPT samples were taken with a safety hammer (ER = 70%): N = 11 at 12 ft and N = 24 at 45 ft. The geologist claims the deeper sand is 'much denser' because its raw N is ~65% higher. The rig used 2 in. OD split spoon and 1.4 in. ID liner-free sampler.

Most nearly, (N1)60 at 45 ft is:

A. 28
B. 18
C. 23
D. 15

Given equations / data

> N60 = (ER/60)·N; CN = 0.77 log10(20/Po), Po in tsf effective; γw = 62.4 pcf

Worked solution

1. Po(12) = 0.551 tsf → (N1)60 = 11×70/60×1.201 = 15.4
2. Po(45) = [6(118) + 39(128−62.4)]/2000 = 1.633 tsf
3. CN = 0.77 log10(20/1.633) = 0.838
4. (N1)60 = 24×70/60×CN = 23.5; after correction the two sands are much closer than raw N suggests

Final answer

23

Why the other values are wrong

- **28** — forgot the overburden correction
- **18** — used total stress in CN
- **15** — reported the 12 ft value

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.1.1–3.8.1.2, p. 124

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